Literature Review

IT 207 Human Intelligence

Human cognition mapping

Selective attention motivates lane and region filters. Working memory motivates a short traffic-history buffer. Learning motivates supervised traffic-forecast models. Decision-making motivates a multi-criteria signal controller. These are implementation analogies, not psychological measures; the course authority cautions against overclaiming intelligence from proxies [1]. Broadbent [2] and Baddeley [3] provide foundational accounts of attention and working memory.

Vehicle detection

YOLO formulates object detection as a single neural-network prediction over an image [4]. This project configures an Ultralytics YOLO11n model for four COCO vehicle classes. The weights are absent, so the configured detector was not run and no detection accuracy is claimed.

Vehicle tracking

ByteTrack associates detection boxes with tracklets, including lower-confidence detections, to reduce identity fragmentation [5]. The project uses the Ultralytics ByteTrack interface. Track quality requires video annotations; none were supplied, so IDF1 and MOTA remain unmeasured.

Traffic control and simulation

The code compares a fixed allocation, count/wait heuristic, and weighted adaptive allocation. SUMO is an open-source microscopic simulator suitable for traffic control experiments [6], but this workspace's network and phase plan were not validated and SUMO was unavailable. Therefore the present comparison uses only the in-project seeded queue simulator.

Reference project and research gap

Prit2341's repository discusses density-based green allocation, startup lag, vehicle category, lane count, and consecutive-signal coordination [7]. Its README's reported throughput gain is unverified by this study. No source code was reused; no repository license file was found during inspection. The next research step is to test corridor coordination in a validated network with multiple intersections, realistic phase conflicts, calibrated vehicle flow, and multi-seed evaluation.

References

[1] IT 207 Human Intelligence Project Details. Supplied course authority PDF, 6 pages. Seven-chapter outline, assessment rubric, and recommended submission package.

[2] Broadbent, D. E. (1958). Perception and Communication. Pergamon Press.

[3] Baddeley, A. (2000). The episodic buffer: a new component of working memory? Trends in Cognitive Sciences, 4(11), 417-423. https://doi.org/10.1016/S1364-6613(00)01538-2

[4] Redmon, J., Divvala, S., Girshick, R., and Farhadi, A. (2016). You Only Look Once: Unified, Real-Time Object Detection. CVPR, 779-788. https://openaccess.thecvf.com/content_cvpr_2016/html/Redmon_You_Only_Look_CVPR_2016_paper.html

[5] Zhang, Y. et al. (2022). ByteTrack: Multi-Object Tracking by Associating Every Detection Box. ECCV. https://www.ecva.net/papers/eccv_2022/papers_ECCV/papers/136820001.pdf

[6] Eclipse SUMO documentation, SUMO at a Glance. https://sumo.dlr.de/docs/SUMO_at_a_Glance.html

[7] Prit2341. Traffic-Management-Using-AI, README and simulation files. https://github.com/Prit2341/Traffic-Management-Using-AI (accessed 2026-11-02). No source code reused; repository-reported 35% result treated as unverified.